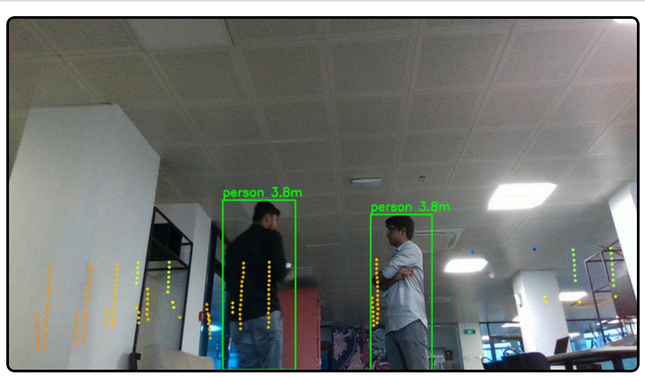
# Fuzzy Engagement-Aware Gesture Command Recognition for a Quadruped Robot in Crowds

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## 01 The problem

Gesture-controlled robots react to any hand that moves. In a crowd, a bystander waving at a friend can stop or start the robot by accident. Current pipelines classify the hand but never ask whose hand it is, or whether that person is even talking to the robot.




*Real Go2 frame: YOLO boxes + LiDAR points (dots) fused into a 3D distance per person. Whose gesture should count?*

**OUR IDEA**

Only trust a gesture from the person who is actually engaging with the robot.

**GESTURE → COMMAND**

Open palm → STOP · Fist → Forward · Call → Backward
One finger → Turn left · Two-up → Turn right
No hand for 0.5 s → robot auto-stops

## 02 Method: two fuzzy layers per person

**Sensors** · RealSense D435i RGB-D + Hesai LiDAR on Unitree Go2 EDU

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**LiDAR–camera fusion tracker**
YOLOv8 + LiDAR → Kalman tracker: ID, distance, approach speed, ± uncertainty

**MediaPipe hand + pose**
Hand landmarks → 18-class MLP gesture + confidence; body pose → gaze toward robot

↓

**Layer 1 · Engagement** — E = f(distance, approach speed, gaze)

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**Layer 2 · Decision** — act = f(E, gesture conf.)

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Command sent only if engaged · /cmd_vel 20 Hz · ROS 2

### From Type-1 to Interval Type-2 fuzzy logic



Type-1: one crisp curve — treats every sensor reading as exact



Interval Type-2: band width set by the tracker's uncertainty — 0.30 m when seen, up to 0.66 m when occluded

Everything runs in ROS 2 inside Docker, so the same code works in simulation and on the robot. We test in a simulated café crowd before moving to the real Go2, and simulated and real data use the same format, so nothing changes between them.

## 03 Results

| 0.864 | 0.91 | ~14 fps | 9 |
|---|---|---|---|
| gesture macro-F1 (±0.018), 71 held-out subjects | STOP recall with safety-biased threshold | hand + pose pipeline, 640×480 | crowd scenarios recorded on the Go2 |

**Bystander false triggers on the robot**

| | Stationary | Walking |
|---|---|---|
| Gesture only | [ — ] | [ — ] |
| + Type-1 engagement | [ — ] | [ — ] |
| **+ Interval Type-2** | **[ — ]** | **[ — ]** |

**KEY FINDING**

[One sentence: how much Interval Type-2 cut bystander false triggers while walking]




*Real Go2 recording: fused LiDAR distance vs independent RealSense depth — median +0.28 m, p90 0.47 m (22 detections)*